

HEMA G

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Summary

Final-year Biomedical Engineering student with a strong foundation in healthcare technology, medical devices, and biomedical instrumentation. Gained hands-on exposure through hospital experience and healthcare-focused projects, with strong communication, teamwork, and problem-solving skills. Able to understand and explain technical concepts clearly and work effectively in team environments. Seeking opportunities to apply technical and interpersonal skills in healthcare technology, service operations, and medical-device environments.

Education

SRM Institute of Science and Technology, Kattankulatham

Bachelor of Technology in Biomedical Engineering

- CGPA: 9.51/10

Tamilnadu, India

Aug 2023 – Present

Holy Cross Matriculation Higher Secondary School

Coursework: Biology, Mathematics, Physics, Chemistry

- Grade: 80.5

Tamilnadu, India

June 2021 – May 2022

Experience

Technical Domain Member, BMESI(Student Chapter)

SRM Institute of Science and Technology

2026 – Present

Tamil Nadu, India

- Contribute to technical initiatives and collaborate with team members on biomedical engineering activities and projects.
- Participate in technical discussions, knowledge-sharing activities, and problem-solving tasks within the team.

Healthcare Domain Head, Aakash Research Labs

SRM University

June 2025 – June 2026

Tamil Nadu, India

- Lead a healthcare project team, managing coordination, planning, and technical execution.
- Facilitating brainstorming sessions and ensuring smooth progress in an innovation-focused environment.

Member, Directorate of Student Affairs (DSA)

SRM University

April 2025 – March 2026

Tamil Nadu, India

- Managed budgeting, fund allocation, and event coordination for student activities including Milan 2025.
- Supported smooth execution of campus events and student programs.

Biomedical Engineer Intern

Sri Narayani Hospital and Research Center

December 2024

Tamil Nadu, India

- Gained hands-on experience in operation, maintenance, and troubleshooting of medical equipment in clinical settings.
- Developed understanding of hospital equipment management and safety standards.

Technical Skills

Programming Languages: Python (Basics)

Technical Domains: IoT Development, AI/ML Fundamentals, Biomedical Instrumentation, Signal Processing, Medical Electronics

Tools & Platforms: Arduino IDE, Wokwi, KiCad, EasyEDA, Tinkercad, Hugging Face Spaces

Core Subjects: Biomedical Instrumentation, Sensors and Transducers, Digital Signal Processing, Embedded Systems

Soft Skills: Leadership, Communication and Presentation, Problem Solving, Teamwork, Adaptability, Project Management, Market Research.

Projects

- Rapid Wound Infection and Healing Analysis Using Affordable Paper-Based Device** 2025 – Present
- Low-cost paper-based colorimetric device for rapid wound infection and healing detection using multi-zone reagent analysis (pH, oxidative stress, inflammatory markers) on Whatman filter paper with wax barriers. Smartphone RGB analysis and UV-Vis spectroscopy used for validation, with web-based remote monitoring integration.
- AI-Based Sepsis ICU Monitoring System** March 2026
- Built an AI-driven ICU monitoring system for early sepsis detection using patient data analytics, deployed on Hugging Face Spaces.
- EEG-EMG Fusion Using Dual-Branch CNN Autoencoder for Anomaly Detection** Jan 2026 – Feb 2026
- Built a dual-branch CNN autoencoder model to fuse EEG and EMG biosignals for anomaly detection in neurological and muscular activity patterns.
- Multi-Sensor Smart Cane for the Visually Impaired** Oct 2025
- Smart assistive cane with ultrasonic, infrared, and rain sensors for real-time obstacle detection, built using Arduino Nano with Embedded C and Tinkercad.
- IoT-Based Pulse Oximeter with Blynk Integration** March 2025
- Portable, low-cost pulse oximeter using MAX30100 and ESP8266 to monitor heart rate (BPM) and SpO₂ in real time, with OLED display and wireless data transfer via the Blynk app.
- Heartbeat Sensor Circuit** March 2025
- Heartbeat sensor circuit using LM358 op-amp and IR LED photodiode to detect pulse rate by measuring blood flow changes at the fingertip, with BPM output and blinking LED pulse indication.
- Solar-Based Mobile Phone Charger** April 2024
- Eco-friendly solar charger that converts sunlight into electricity to charge mobile devices, using a solar panel, rechargeable battery, voltage regulator, and USB module.

Publications

- Bava Prateeksha K S, Hema G**, “EEG–EMG Fusion Using Dual-Branch CNN Autoencoder for Anomaly Detection,” *International Conference on Modern Trends in Engineering, Science, Management, and Technology (MESMT-2026)*, p. 300, Mar. 2026.
- Hema G, Anish T, Karthick Raja, Jaligam Murali Mohan**, “Rapid Analysis of Wound Monitoring for Infection and Healing Analysis using Affordable Device,” *International Conference on Bioengineering (ICB-2026)*, Mar. 2026.

Achievements

- First Prize – Ideathon Competition, ICB 2026:** Awarded First Prize for presenting “*AI Based Sepsis Prediction Using Multimodel ICU Data*” in the Ideathon Competition under the Healthcare and Innovation domain at the International Conference on Bioengineering (ICB 2026), SRM Institute of Science and Technology, March 2026.

Certifications

- AICTE:** Embedded System Developer Virtual Internship
- World Health Organization (Sep 2024):** Medical Equipment Electrical Safety Testing
- COURSERA (Apr 2024):** Crash Course on Python
- MATHWORKS (June 2025):** MATLAB Onramp, Signal Processing Onramp, Image Processing Onramp

Additional Information

- Languages:** Tamil, English
- Programming Languages:** Basics in Python and MATLAB